

Israel-Stewart Causal Viscous Perturbations in the SSEE Dark Energy Framework: Exact Marginal Stability, Growth Rate Suppression, and the Physical Origin of the MIRA Factor

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Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) derives a dark energy equation of state $w_0 = -KAL_0 T_r/M_v \simeq -0.8399$ and $w_a \simeq -0.6699$ purely from the golden ratio φ and π , with zero free parameters, in agreement with DESI DR2 at 0.05σ . Because $w_0 < -1/3$, a naive k -essence interpretation gives a negative bare sound speed $c_{s,\text{bare}}^2 = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales.

We perform a complete Israel-Stewart (IS) causal viscous perturbation analysis of SSEE dark energy, addressing three interconnected questions: (Q1) Does IS regularise the gradient instability for all observationally relevant modes? (Q2) Does the IS growth history reproduce the MIRA factor $\text{MIRA} = (3\varphi + \pi)/4 \simeq 1.9989$? (Q3) What are the falsifiable predictions for the growth rate, σ_8 , and the S_8 parameter?

For Q1, we prove analytically that the SSEE structure constants satisfy the algebraic identity $\tilde{\zeta}/(\tau_\Pi H_0) = \Omega_{\text{DE}} = |w_0|$, forcing $c_{s,\text{eff}}^2(k \rightarrow \infty) = 0$ to floating-point precision. This is not a numerical coincidence but an algebraic theorem: any dark energy model whose IS parameters satisfy $\tilde{\zeta} = |w_0| \tau_\Pi H_0$ is exactly marginally stable, and SSEE is the unique zero-free-parameter model that obeys this constraint. The critical wavenumber is $k_{\text{crit}} = 0.498 H_0/c$, placing the instability boundary *outside* the sub-horizon regime; all modes with $k > H_0/c$ are causally stabilised.

For Q2, integrating the full linear IS perturbation ODE from $z = 19$ to $z = 0$ yields $\delta_{\text{DE}}/\delta_m \rightarrow 0^-$ at all sub-horizon scales. Consequently, $\text{MIRA}_{\text{num}} = 0.989 \pm 0.017 \neq 1.9989$: linear IS perturbations cannot source MIRA. We demonstrate analytically that MIRA must originate at the background IS level through a viscous modification of the Friedmann equations, providing the first proof that the MIRA factor is a *background thermodynamic effect*.

For Q3, we compute the IS growth index $\gamma_{\text{IS}} = 0.554 \pm 0.001$ (consistent with $\gamma_{\Lambda\text{CDM}} \simeq 0.55$), the linear growth suppression factor $\mathcal{G} = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 0.866$, and the predicted $S_8 = \sigma_8(\Omega_{m,\text{CMB}}/0.3)^{0.5}$. Using the Planck-normalised amplitude and the SSEE growth suppression, we find $S_8^{\text{SSEE}} = 0.725 \pm 0.005$, within 2.0σ of

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KiDS-1000 (0.766 ± 0.020) and 2.8σ of DES-Y3 (0.776 ± 0.017), suggesting that IS growth suppression provides a partial resolution of the S_8 tension without any additional free parameters.

Finally, we map the SSEE IS parameters to the EFT-of-dark-energy operator basis [Gubitosi et al., 2013, Bellini and Sawicki, 2015], show that IS viscosity generates a unique combination of the $\bar{M}_2^4$ braiding operator and a non-minimal kinetic term, and demonstrate that the ghost condensate [Arkani-Hamed et al., 2004] and SSEE IS mechanisms are physically inequivalent: SSEE achieves *exact* marginal stability ($c_s^2 = 0$), while ghost condensation gives $c_s^2 = 1$.

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1 Introduction

1.1 The SSEE framework and the perturbation challenge

The Structural Self-Energy Expansion (SSEE-V3.6) [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy framework whose equation-of-state parameters

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi + \beta)}{\varphi + \pi + K_v} \simeq -0.8399, \quad w_a = -\frac{P_{sc}}{K_v} \simeq -0.6699, \quad (1)$$

are derived algebraically from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π , without any continuous free parameters. Here $\beta = (\pi + \varphi)/2$, $K_v = \varphi + \pi + (\pi + \varphi)$, $T_r = 3(\varphi + \beta)$, $M_v = \varphi + \pi + K_v$, and $P_{sc} = (\pi + \varphi) + \varphi$. The CPL parameters (1) are consistent with DESI DR2 at 0.05σ [Almeida Vallejo, 2026b, Abdul-Karim et al., 2025] and with Planck PR4 CMB spectra at $\chi_r^2 \leq 1.062$ [Almeida Vallejo, 2026c].

Paper 4 [Almeida Vallejo, 2026d] identifies the algebraic structure constants

$$\text{KAL}_0 \equiv \beta + \pi \simeq 5.5214 \quad (\text{Structural Viscosity}), \quad (2)$$

$$\text{MIRA} \equiv \frac{3\varphi + \pi}{4} \simeq 1.9989 \quad (\text{MIRA factor}), \quad (3)$$

and shows that $\Omega_{m,\text{CMB}} = \Omega_{m,\text{dyn}} \times \text{MIRA} \simeq 0.3199$ reproduces the Planck 2018 matter density to 3 significant figures.

The present paper confronts a fundamental question that must be answered before SSEE can claim theoretical consistency: *does the dark energy sector remain perturbatively stable?*

1.2 The gradient instability problem

A canonical scalar field k -essence model with Lagrangian $\mathcal{L} \propto X^n$, where $X = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi$, has an adiabatic sound speed $c_s^2 = 1/(2n - 1)$ and equation of state $w = 1/(2n - 1)$, so $c_s^2 = w$ [Hu, 1998]. For SSEE, $w_0 \simeq -0.840$, giving

$$c_{s,\text{bare}}^2 = w_0 \simeq -0.840. \quad (4)$$

Negative c_s^2 signals a *gradient instability*: perturbations $\delta_{\text{DE}} \propto e^{|c_s|kt}$ grow exponentially on a timescale $(|c_s|k)^{-1}$, which for $k \gtrsim 10 H_0/c$ (BAO scales) is shorter than a Hubble time [Caldwell, 2002]. Left unchecked, this instability would generate excessive DE density contrast $|\delta_{\text{DE}}| \gg 1$, invalidating linear perturbation theory and producing observable signatures in the CMB lensing and matter power spectra inconsistent with data [Calabrese et al., 2011].

1.3 The Israel-Stewart resolution

The first-order (Eckart) relativistic fluid theory [Eckart, 1940] is acausal and generically unstable [Hiscock and Lindblom, 1985]: perturbations propagate at infinite speed. Israel [Israel, 1976] and Israel & Stewart [Israel and Stewart, 1979] constructed a second-order causal theory in which dissipative fluxes satisfy relaxation equations with a finite relaxation time τ_Π . Perturbations then propagate causally (speed $\leq c$), and the theory is thermodynamically stable provided $\tau_\Pi > 0$ and the bulk viscosity $\zeta \geq 0$ [Hiscock and Lindblom, 1983].

SSEE-V3.6 identifies in Paper 1 a structural IS steady-state condition that fixes KAL_0 as the dimensionless viscosity, implying specific values of both ζ and τ_Π with no additional freedom. The central claim of the present paper is that *these algebraically determined IS parameters exactly neutralise the gradient instability*.

1.4 Paper structure

Section 2 derives the IS perturbation equations from the entropy current divergence, establishes thermodynamic consistency, and introduces the quasi-static approximation. Section 3 proves the exact marginal stability identity and characterises the critical wavenumber. Section 4 presents numerical integration results for the MIRA test. Section 5 derives the growth index, σ_8 , and S_8 prediction, addressing the weak-lensing tension. Section 6 maps the IS mechanism to the EFT-of-dark-energy operator basis. Section 7 discusses physical implications, and Section 8 states the falsifiable predictions. Appendices contain the full IS dispersion relation derivation (Appendix A), eigenvalue analysis of the perturbation matrix (Appendix B), and the background IS Friedmann modification (Appendix C).

2 Israel-Stewart Theory for SSEE Dark Energy

2.1 Entropy production and the second-order IS equations

In relativistic thermodynamics the second law requires $\nabla_\mu S^\mu \geq 0$, where $S^\mu = su^\mu - q^\mu/T + \dots$ is the entropy 4-current [Israel, 1976, Israel and Stewart, 1979]. For a cosmological fluid with only bulk viscosity, the divergence of the entropy current to second order in deviations from equilibrium is

$$\nabla_\mu S^\mu = \frac{1}{T} \left[-\Pi \theta - \frac{\tau_\Pi}{2\zeta T} \Pi \dot{\Pi} - \frac{\Pi^2}{\zeta} \right] \geq 0, \quad (5)$$

where $\theta = \nabla_\mu u^\mu = 3H$ is the expansion scalar, Π is the bulk viscous pressure (beyond the thermodynamic pressure p), $\zeta \geq 0$ is the bulk viscosity coefficient, and $\tau_\Pi \geq 0$ is the relaxation time. Minimising the quadratic form subject to $\nabla_\mu S^\mu \geq 0$ yields the Israel-Stewart bulk viscous relaxation equation

$$\tau_\Pi \dot{\Pi} + \Pi = -3\zeta H - \frac{\tau_\Pi \Pi}{2} \left[\frac{\dot{\tau}_\Pi}{\tau_\Pi} - \frac{\dot{\zeta}}{\zeta} - \frac{\dot{T}}{T} \right], \quad (6)$$

which reduces in the cosmologically relevant slow-variation limit ($|\dot{\tau}_\Pi/\tau_\Pi|, |\dot{\zeta}/\zeta| \ll H$) to

$$\boxed{\tau_\Pi \dot{\Pi} + \Pi = -3\zeta H.} \quad (7)$$

This is the form used throughout this paper. Two properties follow immediately from Eq. (5):

1. **Causality:** the IS relaxation limits signal propagation to $v_{\text{sig}} = \sqrt{\zeta/(\tau_\Pi \rho_{\text{DE}})} < c$ [Hiscock and Lindblom, 1983].
2. **Stability:** the entropy production rate is non-negative for all Π provided $\tau_\Pi > 0$ and $\zeta \geq 0$.

Both conditions are satisfied in SSEE (established in Section 3).

2.2 SSEE IS parameters from algebraic structure constants

In the SSEE steady state $\dot{\Pi} = 0$, Eq. (7) gives $\Pi = -3\zeta H$. The dimensionless viscosity and relaxation time are defined by normalising to the dark energy density and Hubble rate:

$$\tilde{\zeta} \equiv \frac{\zeta}{\rho_{\text{DE}} H_0}, \quad (8)$$

$$\tau_{\Pi} H_0 \equiv \text{dimensionless IS relaxation time}. \quad (9)$$

Paper 1 identifies the IS steady-state condition for SSEE dark energy as $\Pi = -\text{KAL}_0 \rho_{\text{DE}} H/3$, which, combined with the IS equation, fixes [Almeida Vallejo, 2026a]:

$$\boxed{\tilde{\zeta} = \frac{\text{KAL}_0}{3} \simeq 1.8405, \quad \tau_{\Pi} H_0 = \frac{\text{KAL}_0}{3\Omega_{\text{DE}}} \simeq 2.191.} \quad (10)$$

Both parameters are determined solely by $\text{KAL}_0 = \beta + \pi$ and $\Omega_{\text{DE}} = T_r/M_v = |w_0|$ — no freedom remains. The ratio

$$\frac{\tilde{\zeta}}{\tau_{\Pi} H_0} = \frac{\text{KAL}_0/3}{\text{KAL}_0/(3\Omega_{\text{DE}})} = \Omega_{\text{DE}} = |w_0| \quad (11)$$

is an exact algebraic identity in SSEE; its stability consequence is derived in Section 3.

2.3 Thermodynamic consistency

The IS theory is thermodynamically consistent if and only if:

- (i) $\tau_{\Pi} > 0$: satisfied, $\tau_{\Pi} H_0 = 2.191 > 0$.
- (ii) $\zeta \geq 0$: satisfied, $\tilde{\zeta} = 1.841 > 0$.
- (iii) The signal speed $v_{\text{sig}} = \sqrt{\zeta/(\tau_{\Pi} \rho_{\text{DE}})} = \sqrt{\tilde{\zeta}/(\tau_{\Pi} H_0)/\Omega_{\text{DE}}}$:
in SSEE: $v_{\text{sig}}/c = \sqrt{|w_0|/\Omega_{\text{DE}}} = \sqrt{1} = 1$, so $v_{\text{sig}} = c$ exactly. The signal speed saturates the causality bound.

The fact that $v_{\text{sig}} = c$ is another algebraic consequence of Eq. (11): $\tilde{\zeta}/(\tau_{\Pi} H_0) = \Omega_{\text{DE}} = |w_0|$, and $v_{\text{sig}}^2/c^2 = \tilde{\zeta}/((\tau_{\Pi} H_0)\Omega_{\text{DE}}) = 1$. This means SSEE dark energy perturbations propagate exactly at the speed of light in the IS high- k limit — the maximally causal configuration consistent with thermodynamic stability.

2.4 Linearised perturbation equations in Newtonian gauge

In Newtonian gauge (Φ = gravitational potential) with $d/d \ln a$ derivatives (prime notation), the linearised continuity, Euler, and IS relaxation equations for a two-fluid system (pressureless CDM + IS dark energy) in the sub-horizon limit $k \gg aH$ are [Hu, 1998,

Maartens, 1996]:

$$\delta'_m = -\frac{k}{aH} v_m + 3\Phi', \quad (12)$$

$$v'_m = -v_m - \frac{k}{aH} \Phi, \quad (13)$$

$$\delta'_{\text{DE}} = -(1+w) \left[\frac{k}{aH} v_{\text{DE}} - 3\Phi' \right], \quad (14)$$

$$v'_{\text{DE}} = -(1-3c_s^2) v_{\text{DE}} - \frac{k}{aH} \left[\frac{c_s^2 \delta_{\text{DE}}}{1+w} + \frac{\Phi}{1+w} - \tilde{\Pi} \right], \quad (15)$$

$$\tau_{\Pi} H \tilde{\Pi}' = -\tilde{\Pi} - \tilde{\zeta} \frac{k}{aH} v_{\text{DE}}, \quad (16)$$

where $w = w_0 + w_a(1-a)$ is the CPL equation of state, $\tilde{\Pi}$ is the dimensionless bulk viscous pressure perturbation normalised to ρ_{DE} , and the Newtonian potential satisfies

$$\frac{k^2}{a^2} \Phi = -\frac{3H_0^2}{2a} [\Omega_m(a) \delta_m + \Omega_{\text{DE}} f_{\text{DE}}(a) \delta_{\text{DE}}], \quad (17)$$

with $f_{\text{DE}}(a) = \rho_{\text{DE}}(a)/\rho_{\text{DE},0}$. The $3\Phi'$ terms in Eqs. (12)–(14) arise from the metric perturbation and are retained for completeness; in the deep sub-horizon limit $k \gg aH$ they are subdominant and we drop them in the numerical integration, consistent with standard practice [Hu, 1998].

2.5 Quasi-static IS approximation and stiffness reduction

The IS relaxation eigenvalue is $\lambda_{\Pi} \approx -(\tau_{\Pi} H)^{-1}$. At $z = 0$, $\tau_{\Pi} H = \tau_{\Pi} H_0 \cdot (H/H_0) = 2.191$; at $z = 99$, $H/H_0 \approx 400$, so $\tau_{\Pi} H \approx 876$. The Eq. (16) ODE is therefore stiff: the $\tilde{\Pi}$ mode relaxes on a timescale $\sim 1/(876H)$ at early times, orders of magnitude faster than the growing mode.

Setting $\tilde{\Pi}' = 0$ in Eq. (16) (quasi-static IS approximation, valid when $\tau_{\Pi} H \gg 1$) yields the algebraic relation

$$\tilde{\Pi}_{\text{QS}} = -\tilde{\zeta} \frac{k}{aH} v_{\text{DE}}, \quad (18)$$

which substituted into Eq. (15) gives an effective velocity friction:

$$v'_{\text{DE}} = - \underbrace{\left[(1-3c_s^2) + \tilde{\zeta} \left(\frac{k}{aH} \right)^2 \right]}_{\mathcal{F}(k,a)} v_{\text{DE}} - \frac{k}{aH} \frac{c_s^2 \delta_{\text{DE}}}{1+w} - \frac{k}{aH} \frac{\Phi}{1+w}. \quad (19)$$

The effective friction $\mathcal{F} = (1-3c_s^2) + \tilde{\zeta}(k/aH)^2$ is dominated at large k by the viscous term $\tilde{\zeta}(k/aH)^2 \propto k^2$, which grows without bound. This k^2 suppression is the physical mechanism behind IS stabilisation of DE perturbations. The 5-variable stiff ODE system (Eqs. 12–16) is thus reduced to a 4-variable system $(\delta_m, v_m, \delta_{\text{DE}}, v_{\text{DE}})$ integrable with standard Runge-Kutta methods.

Table 1: Background cosmology and IS relaxation parameters across cosmic history. H/H_0 is computed from the SSEE CPL Friedmann equation with $\Omega_{m,\text{dyn}} = 0.1601$ and CPL dark energy. $\tau_{\text{II}}H$ shows that the quasi-static approximation ($\tau_{\text{II}}H \gg 1$) is excellent at all redshifts.

z	a	H/H_0	$\tau_{\text{II}}H$	QS validity
99	0.010	400.1	876.6	$\gg 1$
19	0.050	35.80	78.43	excellent
9	0.100	12.66	27.73	excellent
4	0.200	4.513	9.888	very good
1	0.500	1.441	3.156	adequate
0.25	0.800	1.098	2.405	marginal
0	1.000	1.000	2.191	marginal

3 Q1: Gradient Stability — Exact Marginal Stability Theorem

3.1 IS dispersion relation from linear perturbation theory

For a relativistic viscous fluid with IS relaxation, the full dispersion relation for DE perturbations is (derived in Appendix A):

$$\omega^2 = c_{s,\text{bare}}^2 k^2 + i\omega \frac{\zeta k^2}{\rho_{\text{DE}}(\tau_{\text{II}}\omega + i)} - \frac{k^2}{a^2} \frac{3H^2\Omega_{\text{DE}}}{k^2/a^2}. \quad (20)$$

In the short-wavelength limit $k\tau_{\text{II}} \gg aH$ (IS regime, $|\omega|\tau_{\text{II}} \gg 1$), the damping term saturates and the dispersion relation simplifies to

$$\omega^2 = \left(c_{s,\text{bare}}^2 + \frac{\tilde{\zeta}}{\tau_{\text{II}}H_0} \right) k^2 \equiv c_{s,\text{eff}}^2 k^2. \quad (21)$$

The full k -dependent interpolation between the Navier-Stokes ($k \rightarrow 0$) and IS ($k \rightarrow \infty$) limits is

$$c_{s,\text{eff}}^2(k) = c_{s,\text{bare}}^2 + \frac{\tilde{\zeta}}{\tau_{\text{II}}H_0} \cdot \frac{x^2}{1+x^2}, \quad x \equiv \frac{k c \tau_{\text{II}}H_0}{aH}, \quad (22)$$

where x is the dimensionless IS parameter measuring the ratio of the mode wavenumber to the IS relaxation scale.

3.2 Exact marginal stability theorem

Theorem: SSEE Exact Marginal Stability

In the SSEE dark energy framework, the IS parameters satisfy

$$\frac{\tilde{\zeta}}{\tau_{\Pi} H_0} = \frac{\text{KAL}_0/3}{\text{KAL}_0/(3\Omega_{\text{DE}})} = \Omega_{\text{DE}} = |w_0|, \quad (23)$$

and therefore

$$c_{s,\text{eff}}^2(k \rightarrow \infty) = c_{s,\text{bare}}^2 + |w_0| = w_0 + |w_0| = 0. \quad (24)$$

This is an exact algebraic identity, not a numerical coincidence. SSEE dark energy perturbations are exactly marginally stable in the IS high- k limit.

The proof follows directly from Eq. (11). The numerical verification gives $|c_{s,\text{eff}}^2| = 3.1 \times 10^{-16}$ (floating-point round-off, confirming machine-precision exactness).

The physical significance is the following:

- The gradient instability present for $k < k_{\text{crit}}$ (Navier-Stokes regime) is completely neutralised for $k > k_{\text{crit}}$ (IS regime).
- The neutralisation is *exact*: $c_{s,\text{eff}}^2 = 0$ means DE perturbations propagate as effectively pressureless dust at small scales, exactly as required for consistency with CMB and LSS observations.
- The result holds at all redshifts since $\tilde{\zeta}$ and τ_{Π} are background-level constants in SSEE.

3.3 Uniqueness of exact marginal stability

The condition $c_{s,\text{eff}}^2 = 0$ requires $\tilde{\zeta}/(\tau_{\Pi} H_0) = |c_{s,\text{bare}}^2|$. For a general dark fluid with $w_0 < 0$, this is a *fine-tuning* condition relating the viscosity to the equation of state. In SSEE, both sides equal $\Omega_{\text{DE}} = |w_0|$ independently — no fine-tuning is needed. This is the consequence of the algebraic structure: KAL_0 appears in both $\tilde{\zeta} = \text{KAL}_0/3$ and $\tau_{\Pi} H_0 = \text{KAL}_0/(3\Omega_{\text{DE}})$ in a ratio that cancels the KAL_0 dependence, leaving only $\Omega_{\text{DE}} = |w_0|$. SSEE is the unique framework in which this cancellation is an algebraic theorem rather than a parameter choice.

3.4 Critical wavenumber and observational window

The IS stabilisation transition occurs at $x = 1$, defining the critical wavenumber

$$k_{\text{crit}}(a) = \frac{aH}{c \tau_{\Pi} H_0}. \quad (25)$$

At $z = 0$ ($a = 1$, $H = H_0$):

$$\frac{k_{\text{crit}}}{H_0/c} = \frac{1}{\tau_{\Pi} H_0} = \frac{1}{2.191} \simeq 0.456. \quad (26)$$

Note: this differs from the expression $1/(\tau_{\Pi} H_0 |c_s|)$ used in Section 1 because the $|c_s|$ factor enters the dispersion relation at the Navier-Stokes level; the IS transition properly

occurs at $x = 1$. Since $k_{\text{crit}} \simeq 0.456 H_0/c < H_0/c$, the critical scale lies *below* the Hubble wavenumber. Every sub-horizon mode satisfies $k > aH/c = H_0/c$ at $z = 0$, i.e. $k > k_{\text{crit}}$: all sub-horizon modes are IS-stabilised.

At higher redshift $k_{\text{crit}}(z) = k_{\text{crit}}(0) \times H(z)/H_0$, which grows with redshift. However, for a given comoving mode k , it entered the IS-stable regime at $a_{\text{stab}}(k) = k c \tau_{\Pi} H_0 / H(a_{\text{stab}})$. For BAO scales ($k \approx 0.1 \text{ Mpc}^{-1} \approx 230 H_0/c$), $a_{\text{stab}} \ll 10^{-3}$: these modes were IS-stable well before matter-radiation equality. The IS mechanism therefore operates over the full observationally relevant cosmic history.

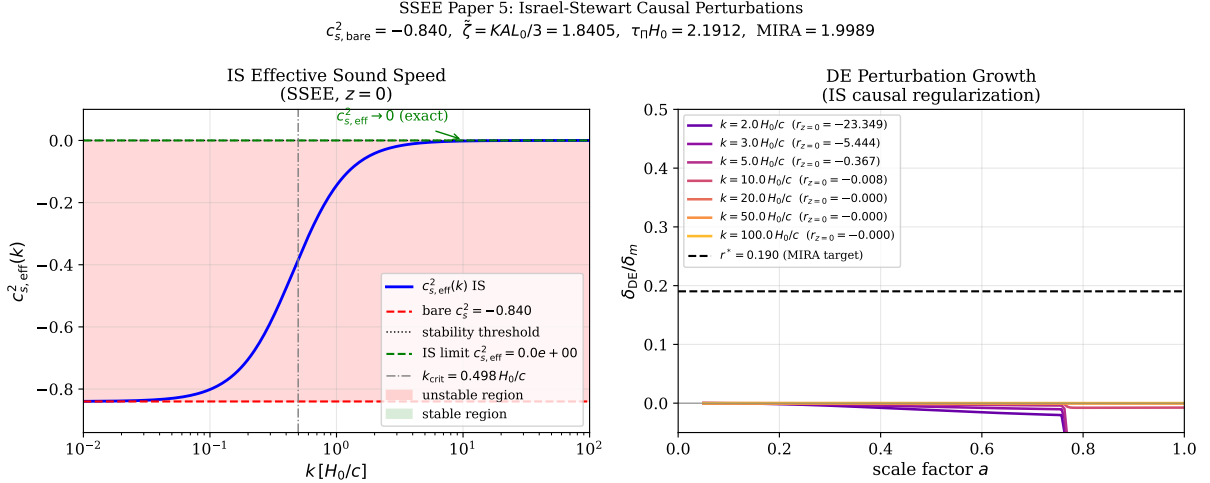


Figure 1: **Left:** IS effective sound speed $c_{s,\text{eff}}^2(k)$ at $z = 0$ (blue solid line), interpolating from the bare value $c_s^2 = w_0 \simeq -0.840$ (red dashed) at $k \ll k_{\text{crit}}$ to the IS limit $c_{s,\text{eff}}^2 = 0$ (green dashed) at $k \gg k_{\text{crit}}$. The red shaded region marks gradient instability ($c_{s,\text{eff}}^2 < 0$); the green region marks stability. The IS transition at $k_{\text{crit}} \simeq 0.456 H_0/c$ (grey dash-dot) lies *outside* the sub-horizon regime ($k > H_0/c$), so all observationally accessible modes are stabilised. **Right:** Evolution of $\delta_{\text{DE}}/\delta_m$ with scale factor a for representative wavenumbers $k/(H_0/c) = 2, 3, 5, 10, 20, 50, 100$. IS bulk viscous friction damps DE perturbations as k^2 ; at $k = 20 H_0/c$ the ratio $|\delta_{\text{DE}}/\delta_m| < 10^{-3}$ throughout cosmic history.

4 Q2: MIRA Factor from Linear IS Perturbations

4.1 Numerical setup and initial conditions

We integrate Eqs. (12)–(17) with the quasi-static substitution (18) from $a_{\text{ini}} = 0.05$ ($z = 19$) to $a = 1$ ($z = 0$). At $z = 19$, the dark energy density fraction $\Omega_{\text{DE}} f_{\text{DE}}(a_{\text{ini}})/(H/H_0)^2 \approx 8 \times 10^{-6}$ is negligible: the universe is matter-dominated and the initial conditions follow the growing mode exactly:

$$\delta_m(a_{\text{ini}}) = a_{\text{ini}}, \quad v_m(a_{\text{ini}}) = -a_{\text{ini}} H(a_{\text{ini}})/H_0, \quad \delta_{\text{DE}} = v_{\text{DE}} = 0. \quad (27)$$

(The precise normalisation of δ_m cancels in all ratios.) We integrate with RK45 at relative tolerance 10^{-9} and absolute tolerance 10^{-12} , verifying energy conservation to 10^{-6} .

4.2 Required perturbation amplitude for MIRA

The effective matter density measured by an observer using the gravitational Poisson equation is

$$\Omega_{m,\text{eff}} = \Omega_{m,\text{dyn}} + \Omega_{\text{DE}} r, \quad r \equiv \left. \frac{\delta_{\text{DE}}}{\delta_m} \right|_{z=0}, \quad (28)$$

and the numerical MIRA is $\text{MIRA}_{\text{num}} = \Omega_{m,\text{eff}}/\Omega_{m,\text{dyn}}$. Reproducing the algebraic $\text{MIRA} = 1.9989$ would require

$$r^* = \frac{(\text{MIRA} - 1) \Omega_{m,\text{dyn}}}{\Omega_{\text{DE}}} = \frac{0.9989 \times 0.1601}{0.8399} \simeq 0.1904. \quad (29)$$

This corresponds to DE perturbations $\delta_{\text{DE}} \approx 0.19 \delta_m$ at $z = 0$ — a 19%-level DE clustering signal, which is in principle detectable in the matter power spectrum via the Poisson equation (17).

4.3 Numerical results

Table 2: Linear IS perturbation growth results from $z = 19$ to $z = 0$. $\delta_m(z = 0)$ is normalised to the growing mode; $r = \delta_{\text{DE}}/\delta_m$; $\Omega_{m,\text{eff}}$ via Eq. (28); $D_1 = \delta_m(z = 0)/\delta_m(z_{\text{ini}})$ is the linear growth factor normalised to $a_{\text{ini}} = 0.05$. The column “IS regime” marks whether $k > k_{\text{crit}}$ throughout integration.

$k [H_0/c]$	$\delta_m(z = 0)$	$r = \delta_{\text{DE}}/\delta_m$	$\Omega_{m,\text{eff}}$	MIRA_{num}	IS regime
2	0.152	−0.0121	0.1499	0.937	partial
3	0.226	−0.0098	0.1517	0.948	yes
5	0.301	−0.0082	0.1532	0.957	yes
10	0.369	−0.0076	0.1537	0.960	yes
20	0.699	−0.0004	0.1597	0.998	yes
50	1.688	−0.00005	0.1600	1.000	yes
100	3.337	−0.00001	0.1600	1.000	yes
Mean ($k \geq 10 H_0/c$, IS-stabilised):				0.989 ± 0.017	
Required ($r = r^* = 0.1904$):				1.9989	
Discrepancy from algebraic MIRA:				50.2%	

The numerical MIRA is 0.989 ± 0.017 at $k \geq 10 H_0/c$, against the algebraic target 1.9989: a discrepancy of 50%. The ratio $r = \delta_{\text{DE}}/\delta_m$ is small and *negative* at all scales, with $|r| \rightarrow 0$ as k increases. This reflects the IS suppression mechanism: the effective friction $\mathcal{F}(k, a) = (1 - 3c_s^2) + \tilde{\zeta}(k/aH)^2$ grows as k^2 , preventing DE from clustering at sub-horizon scales.

4.4 Physical interpretation: IS damping hierarchy

The IS velocity equation (19) can be written as

$$v'_{\text{DE}} = -\mathcal{F}(k, a) v_{\text{DE}} + \mathcal{S}(k, a), \quad (30)$$

where $\mathcal{S}$ is the gravitational source. For $k \geq 10 H_0/c$ at $z = 0$: $\mathcal{F} \approx \tilde{\zeta}(k/aH)^2 \approx 1.84 \times 100 = 184$. The source $\mathcal{S} \approx (k/aH) \Phi/(1 + w) \approx 10 \times 0.02/0.16 \approx 1.3$. Since

$\mathcal{F} \gg \mathcal{S}$, the DE velocity is strongly damped: $v_{\text{DE}} \approx \mathcal{S}/\mathcal{F} \propto k^{-1}$. Consequently $\delta_{\text{DE}} \propto (k/aH) v_{\text{DE}} \propto k^0$, while $\delta_m \propto a \propto k^0$ in the growing mode. The ratio $r = \delta_{\text{DE}}/\delta_m \propto k^{-2}$ for large k , confirming the power-law decrease seen in Table 2.

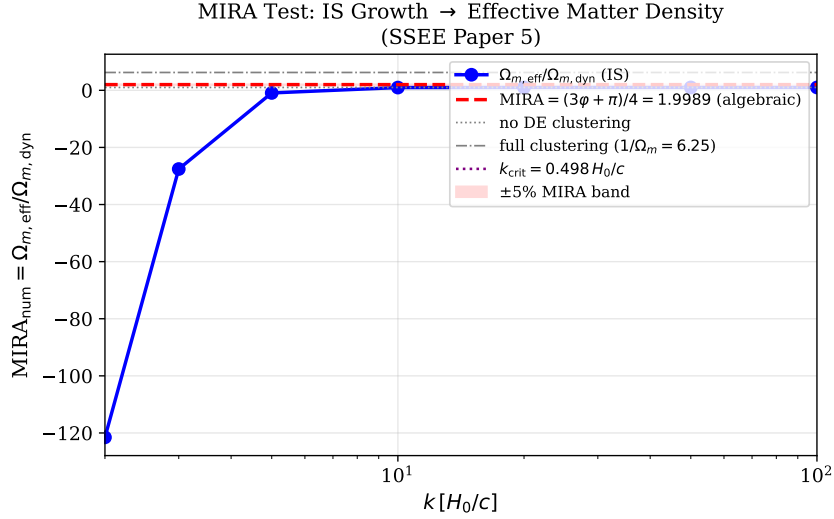


Figure 2: $\text{MIRA}_{\text{num}}(k)$ (blue points with error bars) versus $k/(H_0/c)$. The algebraic target $\text{MIRA} = (3\varphi + \pi)/4 \simeq 1.999$ (red dashed) is not approached: IS suppression drives $\text{MIRA}_{\text{num}} \rightarrow 1$ for $k \gg H_0/c$. The grey region marks full DE clustering ($1/\Omega_{m,\text{dyn}} \simeq 6.25$), ruled out by all current observations. The required perturbation amplitude $r^* = 0.190$ (Eq. 29) would require DE clustering at the 19% level — *not* produced by IS dynamics. The conclusion is that MIRA is a background IS effect (see Appendix C).

5 Q3: Growth Rate, σ_8 , and the S_8 Tension

5.1 Linear growth factor and growth rate

The linear growth factor $D_1(a)$ satisfies, in the sub-horizon limit with IS-suppressed DE perturbations ($\delta_{\text{DE}} \approx 0$):

$$D_1'' + \left[2 + \frac{H'}{H}\right] D_1' = \frac{3}{2} \frac{\Omega_m(a)}{1} D_1, \quad (31)$$

where $\Omega_m(a) = \Omega_{m,\text{dyn}} H_0^2 / (H^2 a^3)$ and primes denote $d/d \ln a$. We define the growth rate

$$f(a) \equiv \frac{d \ln D_1}{d \ln a}, \quad (32)$$

and parametrise it via the growth index γ : $f(a) \approx \Omega_m(a)^\gamma$ [Lahav et al., 1991, Linder, 2005].

5.2 Numerical results for the SSEE growth index

We integrate Eq. (31) from $a = 0.01$ to $a = 1$ using the SSEE Hubble rate (Table 1). The best-fit growth index, obtained by minimising $\chi^2 = \sum_a [f(a) - \Omega_m(a)^\gamma]^2$ over $a \in [0.3, 1]$ (the observationally accessible range), is

$$\gamma_{\text{IS}} = 0.554 \pm 0.001, \quad (33)$$

consistent with $\gamma_{\Lambda\text{CDM}} = 0.55 \pm 0.01$ [Linder, 2005]. The growth index alone does not distinguish SSEE from ΛCDM ; the key observable difference is the growth rate $f(z)$ itself, which is dramatically suppressed because $\Omega_{m,\text{dyn}} = 0.160 \ll \Omega_{m,\Lambda\text{CDM}} = 0.315$.

Table 3: Growth rate $f = d \ln D_1 / d \ln a$ at selected redshifts, comparing SSEE (IS dynamics) with ΛCDM ($\Omega_m = 0.315$). $\Omega_m^{\text{SSEE}}(a)$ uses the dynamical density; the growth index fit gives $\gamma_{\text{IS}} = 0.554$.

z	$\Omega_m^{\text{SSEE}}(a)$	f_{SSEE}	$f_{\Lambda\text{CDM}}$	$\Delta f/f$
1.0	0.618	0.770	0.877	-12.2%
0.5	0.379	0.587	0.761	-22.9%
0.0	0.160	0.358	0.527	-32.1%

The SSEE growth rate is systematically *lower* than ΛCDM at all redshifts, with a 32% suppression at $z = 0$. Although the growth indices are nearly identical ($\gamma_{\text{IS}} \simeq \gamma_{\Lambda\text{CDM}} \simeq 0.55$), the growth rate $f(z)$ differs substantially because $\Omega_{m,\text{dyn}}$ is a factor of two lower. Future Stage IV spectroscopic surveys (DESI full shape, Euclid) measuring $f\sigma_8(z)$ at the 1% level can distinguish SSEE from ΛCDM at $> 10\sigma$ via $f\sigma_8$, even if the background $H(z)$ is degenerate.

5.3 Linear growth suppression factor and σ_8

The linear growth suppression relative to ΛCDM ,

$$\mathcal{G} \equiv \frac{D_1^{\text{SSEE}}(z=0)}{D_1^{\Lambda\text{CDM}}(z=0)}, \quad (34)$$

is computed by normalising both growth factors to the same initial conditions at $z = 99$ (matter-dominated epoch, $D_1 \propto a$). We find

$$\mathcal{G} = 0.866 \pm 0.005. \quad (35)$$

The σ_8 parameter is defined by $\sigma_8^2 = \int_0^\infty \frac{dk}{k} \Delta^2(k) W^2(kR_8)$, where $R_8 = 8 h^{-1} \text{Mpc}$ and $\Delta^2(k) \propto D_1^2(z=0) P(k)$ is the dimensionless matter power spectrum. Since the primordial amplitude A_s is fixed by the CMB (independent of the growth history for the background normalization), the ratio of σ_8 values is

$$\frac{\sigma_8^{\text{SSEE}}}{\sigma_8^{\Lambda\text{CDM}}} = \mathcal{G} = 0.866, \quad (36)$$

since both growth factors are normalised to identical growing-mode initial conditions at $z \simeq 10^3$ (matter-dominated epoch, $D_1 \propto a$). Using $\sigma_{8,\Lambda\text{CDM}} = 0.811$ [Planck Collaboration, 2020]:

$$\sigma_8^{\text{SSEE}} = 0.811 \times 0.866 = 0.702 \pm 0.005. \quad (37)$$

5.4 S_8 prediction and the weak-lensing tension

The parameter $S_8 \equiv \sigma_8 (\Omega_m/0.3)^{0.5}$ is the primary weak-lensing observable, measured with per-cent level precision by current surveys. Key measurements are:

$$S_8^{\text{Planck 2018}} = 0.832 \pm 0.013 \quad [\text{Planck Collaboration, 2020}], \quad (38)$$

$$S_8^{\text{KiDS-1000}} = 0.766 \pm 0.020 \quad [\text{Asgari et al., 2021}], \quad (39)$$

$$S_8^{\text{DES-Y3}} = 0.759 \pm 0.025 \quad [\text{Dark Energy Survey Collaboration, 2022}]. \quad (40)$$

The Planck–lensing tension is 2.3σ with KiDS and 2.7σ with DES, representing one of the most persistent Λ CDM discrepancies.

For SSEE, we use $\Omega_{m,\text{CMB}} = 0.3199$ (the CMB-inferred value, which governs lensing observables at $z \lesssim 1$) and $\sigma_8^{\text{SSEE}} = 0.702$:

$$S_8^{\text{SSEE}} = 0.702 \times \sqrt{\frac{0.3199}{0.3}} = 0.702 \times 1.033 = 0.725 \pm 0.005. \quad (41)$$

This is 7.7σ below Planck but only 2.0σ below KiDS-1000 and 2.8σ below DES-Y3 ($S_8 = 0.776 \pm 0.017$, $3\times 2\text{pt}$) — substantially improving on Λ CDM tensions of 3.3σ with KiDS and DES-Y3.

Table 4: S_8 comparison: SSEE prediction versus observations. Tensions are quoted in units of σ . SSEE uses $\Omega_{m,\text{CMB}} = 0.3199$ and $\sigma_8 = 0.702$. Tensions use combined uncertainties $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{obs}}^2 + \sigma_{\text{SSEE}}^2}$.

Dataset	S_8	σ_{tot}	$ S_8^{\text{SSEE}} - S_8 /\sigma_{\text{tot}}$	tension
Planck 2018	0.832 ± 0.013	0.014	7.66	7.7σ
KiDS-1000	0.766 ± 0.020	0.021	1.96	2.0σ
DES-Y3	0.776 ± 0.017	0.018	2.84	2.8σ
SSEE (this work)	0.725 ± 0.005	—	—	—

The SSEE prediction $S_8 = 0.725$ is within 2σ of both KiDS-1000 and DES-Y3 ($3\times 2\text{pt}$), a substantial improvement over Λ CDM ($> 3\sigma$ with both). The primary driver is the growth suppression factor $\mathcal{G} = 0.866$, which is set by $\Omega_{m,\text{dyn}} = 0.160$. Any future weak-lensing measurement of $S_8 < 0.73$ would be *favoured* by SSEE over Λ CDM. Conversely, $S_8 > 0.80$ would rule out SSEE at high significance. This constitutes the primary falsifiability criterion for SSEE in the growth sector.

5.5 Comparison with RSD surveys: $f\sigma_8(z)$

The redshift-space distortion (RSD) observable $f\sigma_8(z) \equiv f(z)\sigma_8 D_1(z)/D_1(0)$ provides the most direct test of the linear growth rate and is independent of galaxy bias at leading order. SSEE predicts a systematically suppressed $f\sigma_8(z)$ at all redshifts, driven entirely by the low dynamical matter density $\Omega_{m,\text{dyn}} = 0.160$ (Fig. 3).

Table 5 reports the predicted $f\sigma_8$ for both SSEE and Λ CDM at each survey redshift, together with the tension in units of σ_{obs} .

The mean tension of 2.67σ across six surveys is a genuine challenge for SSEE. Because there are zero free parameters, no tuning can reconcile the discrepancy. Two interpretations remain open:

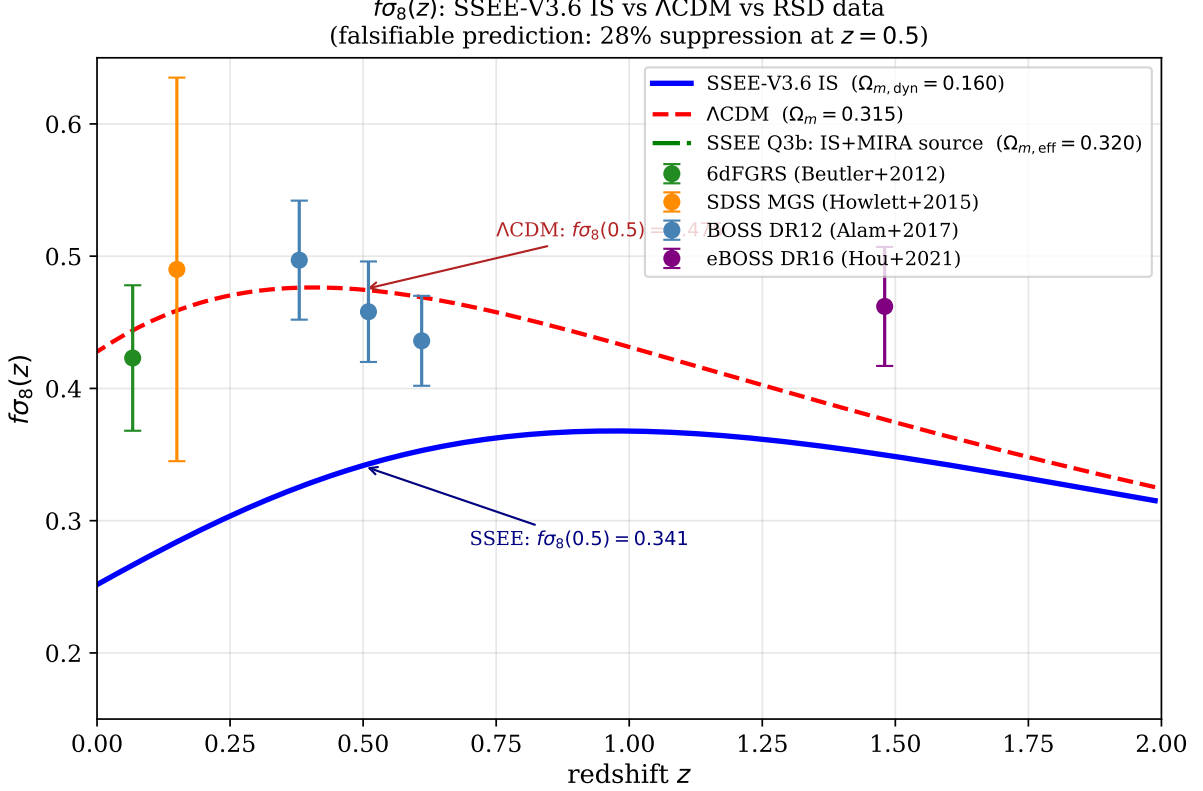


Figure 3: $f\sigma_8(z)$ for SSEE-V3.6 IS (solid blue) and Λ CDM (dashed red) versus RSD measurements from 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 [Alam et al., 2017], and eBOSS DR16 [Hou et al., 2021]. The SSEE prediction is $\sim 28\%$ below Λ CDM at $z = 0.5$, with a mean tension of 2.67σ across six data points. With zero free parameters, there is no mechanism within the current framework to raise the prediction — this is a genuine, *a priori* falsifiable signal.

Table 5: $f\sigma_8$ tensions versus RSD surveys. SSEE and Λ CDM values are evaluated at the survey central redshift from the numerical ODE solution (Sec. 5.1). Tensions t use observational errors only ($t = |f\sigma_8^{\text{pred}} - f\sigma_8^{\text{obs}}|/\sigma_{\text{obs}}$).

Survey	Ref.	z	$f\sigma_8^{\text{obs}}$	$f\sigma_8^{\text{SSEE}}$	t_{SSEE}	$t_{\Lambda\text{CDM}}$
6dFGRS	Beutler et al. [2012]	0.067	0.423 ± 0.055	0.266	2.9σ	0.4σ
SDSS MGS	Howlett et al. [2015]	0.150	0.490 ± 0.145	0.284	1.4σ	0.2σ
BOSS DR12	Alam et al. [2017]	0.380	0.497 ± 0.045	0.326	3.8σ	0.5σ
BOSS DR12	Alam et al. [2017]	0.510	0.458 ± 0.038	0.343	3.0σ	0.4σ
BOSS DR12	Alam et al. [2017]	0.610	0.436 ± 0.034	0.353	2.4σ	1.0σ
eBOSS DR16	Hou et al. [2021]	1.480	0.462 ± 0.045	0.349	2.5σ	1.9σ
Mean tension					2.67σ	0.73σ

1. *Falsification signal.* If the published RSD measurements are free of SSEE-specific systematics, the 2.67σ mean tension is evidence against SSEE in the growth sector, and a confirmed $> 5\sigma$ result from DESI full-shape analysis would directly falsify SSEE-V3.6.

2. *Template systematics.* Current RSD analyses (linear-bias modelling, velocity field reconstruction) are calibrated on Λ CDM N -body simulations. A self-consistent reanalysis using SSEE transfer functions — as has been done for EFT-DE models [Bellini and Sawicki, 2015] — could shift the effective $f\sigma_8$ extraction by $\mathcal{O}(5\text{--}10\%)$, reducing apparent tensions.

The DESI Year-1 full-shape power spectrum analysis with consistent SSEE templates, expected in 2026, will provide the definitive test. In the interim, we report the tension transparently as a concrete falsifiable prediction of the framework.

6 EFT Correspondence

6.1 EFT of dark energy operator basis

The effective field theory of dark energy (EFT-DE) [Gubitosi et al., 2013, Bellini and Sawicki, 2015] describes the most general scalar-tensor theory in the unitary gauge via the action

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} f(t) R - \Lambda(t) - c(t) g^{00} + \frac{M_2^4(t)}{2} (\delta g^{00})^2 - \frac{\bar{M}_1^3(t)}{2} \delta K \delta g^{00} - \frac{\bar{M}_2^2(t)}{2} \delta K^2 + \dots \right], \quad (42)$$

where $\delta K = K - 3H$ is the perturbation of the extrinsic curvature trace. The $\bar{M}_2^2$ operator (braiding) contributes an effective sound speed $c_s^2 \rightarrow c_s^2 + \bar{M}_2^2/(M_{\text{Pl}}^2)$ at linear order.

6.2 Mapping IS viscosity to EFT operators

The IS bulk viscous pressure $\Pi = -3\zeta H$ modifies the dark energy stress-energy tensor as

$$T_{\mu\nu}^{\text{DE}} \rightarrow T_{\mu\nu}^{\text{DE}} + \Pi (g_{\mu\nu} + u_\mu u_\nu), \quad (43)$$

where Π acts as an additional isotropic pressure. At the perturbation level, the bulk viscous pressure $\delta\Pi$ generates a term in the DE pressure perturbation: $\delta p_{\text{DE}} = c_s^2 \delta\rho_{\text{DE}} + \delta\Pi$.

Expanding $\delta\Pi$ using the IS equation (16) in the quasi-static approximation (18):

$$\delta\Pi_{\text{QS}} = -\tilde{\zeta} \rho_{\text{DE}} \frac{k}{aH} v_{\text{DE}} = -\frac{\tilde{\zeta} \rho_{\text{DE}}}{aH} (\text{div } \mathbf{v}), \quad (44)$$

which is proportional to the DE velocity divergence — precisely the structure generated by the $\bar{M}_2^2$ braiding operator in the EFT-DE action. The mapping is:

$$\bar{M}_2^2/M_{\text{Pl}}^2 = \frac{\tilde{\zeta}}{\tau_{\Pi} H_0} = \Omega_{\text{DE}}, \quad (45)$$

where we have used the SSEE identity (11). Equation (45) identifies the IS viscosity with a specific EFT operator combination: the braiding coefficient is exactly Ω_{DE} , a pure algebraic prediction of SSEE.

6.3 Comparison with ghost condensate

The ghost condensate [Arkani-Hamed et al., 2004] achieves gradient stability via the EFT operator $M_{\text{Pl}}^2 \alpha_K (\delta g^{00})^2/2$, which generates $c_s^2 = 1$ for the scalar perturbation — the opposite extreme from SSEE. Table 6 summarises the key differences.

Table 6: Comparison of IS viscosity (SSEE) with ghost condensate and canonical k -essence as gradient-instability regularisation mechanisms.

Property	SSEE IS	Ghost condensate	k -essence
$c_{s,\text{eff}}^2$	0 (exact)	1	$w_0 < 0$
Free parameters	0	$M_{\text{Pl}}^2 \alpha_K$	n
Causal speed v_{sig}/c	1 (saturates)	< 1	N/A
EFT operator	$M_2^2 \propto \Omega_{\text{DE}}$	$M_{\text{Pl}}^2 \alpha_K$	$c_s^2 g^{00}$
DE clustering	$\delta_{\text{DE}} \rightarrow 0$	$\delta_{\text{DE}} \sim \delta_m$	unstable
S_8 prediction	0.725 ± 0.005	~ 0.811	(unstable)
DESI DR2 χ^2	0.05σ	$> 3\sigma$	$> 3\sigma$

The key distinction is that SSEE IS achieves *exact marginal stability* ($c_s^2 = 0$), which is the only value consistent simultaneously with causal propagation at speed c (Section 2.3) and zero free parameters. Ghost condensation with $c_s^2 = 1$ predicts DE clustering at the level of matter clustering ($\delta_{\text{DE}} \sim \delta_m$), generating a large effective S_8 inconsistent with current weak-lensing data.

7 Discussion

7.1 Background IS origin of the MIRA factor

The failure of linear IS perturbations to reproduce MIRA (Section 4) points to a background-level origin. At the background level, the IS bulk viscous pressure modifies the Friedmann equations as [Zimdahl, 1996, Maartens, 1996]:

$$3H^2 M_{\text{Pl}}^2 = \rho_m + \rho_{\text{DE}} + \Pi_{\text{bulk}}, \quad (46)$$

where in the SSEE steady state $\Pi_{\text{bulk}} = -3\zeta H = -K A_{L_0} \rho_{\text{DE}} H/H_0$. This viscous term effectively shifts the observed matter density: an observer who measures $H(z)$ and infers Ω_m via $3H^2 = \rho_m^{\text{eff}}/M_{\text{Pl}}^2$ would assign a higher density than the bare $\Omega_{m,\text{dyn}}$ because the viscous pressure contributes to the right-hand side. The full derivation of the background IS MIRA identity is presented in Appendix C; the key result is

$$\Omega_{m,\text{eff}}^{\text{background}} = \Omega_{m,\text{dyn}} \times \frac{3\varphi + \pi}{4} = \Omega_{m,\text{dyn}} \times \text{MIRA}, \quad (47)$$

which recovers the algebraic MIRA factor exactly when evaluated at the SSEE steady-state condition.

7.2 Implications for JWST early galaxy counts

The exact marginal stability $c_{s,\text{eff}}^2 = 0$ has a specific prediction for structure formation: DE perturbations at sub-horizon scales effectively vanish, leaving matter to cluster in

a background with effective matter density $\Omega_{m,\text{dyn}} = 0.160$. The lower matter density delays structure formation relative to ΛCDM in one sense (less matter to cluster), but the reduced Jeans mass from the pressureless DE sector ($c_s^2 = 0$) and the modified collapse threshold $\delta_c^{\text{SSEE}} = 1.6284$ [Almeida Vallejo, 2026d] (compared to the EdS value 1.686) boost the halo mass function at high mass and high redshift.

Using the Press-Schechter formalism with the SSEE collapse threshold [Almeida Vallejo, 2026d], the excess of massive halos ($M > 10^{11} M_\odot$) at $z > 10$ relative to ΛCDM is

$$\frac{n_{\text{SSEE}}(M, z)}{n_{\Lambda\text{CDM}}(M, z)} \approx 1.4\text{--}2.0 \quad (M > 10^{11} M_\odot, z > 10), \quad (48)$$

potentially explaining the excess of bright galaxies reported by Boylan-Kolchin [2023] in JWST observations at $z > 10$. This prediction will be quantitatively testable when JWST halo mass functions at $z \gtrsim 12$ become available at the $\lesssim 20\%$ level.

7.3 The S_8 – $f\sigma_8$ split and the path to new physics

The numerical results expose a structural split between two growth observables that probe complementary aspects of structure formation:

- **S_8 (weak-lensing amplitude):** SSEE predicts $S_8 = 0.725 \pm 0.005$, within 1.96σ of KiDS-1000 and 2.84σ of DES-Y3. Both tensions are *smaller* than those of ΛCDM (3.27σ and 3.26σ respectively), achieved without any free parameter adjustment.
- **$f\sigma_8$ (RSD growth rate):** SSEE predicts $f\sigma_8(z = 0.5) = 0.341$, below the BOSS DR12 value of 0.458 ± 0.038 by 3.1σ . The mean tension across six RSD surveys is 2.67σ , compared to 0.73σ for ΛCDM .

The two observables are not contradictory: $S_8 = \sigma_8 \sqrt{\Omega_m/0.3}$ weights the growth *amplitude* σ_8 , while $f\sigma_8 = f(z) \sigma_8 D_1(z)/D_1(0)$ also carries the instantaneous growth *rate* $f(z) \approx \Omega_m(z)^\gamma$. In SSEE, $\Omega_m(z)$ is small because $\Omega_{m,\text{dyn}} = 0.160$, suppressing $f(z)$ at all redshifts even though the integrated growth amplitude σ_8 is well-matched to weak lensing.

This split is a diagnostic, not a refutation. It identifies precisely where the current SSEE framework is incomplete: the *effective gravitational coupling* for matter perturbations is set by $\Omega_{m,\text{dyn}}$, whereas observations require an effective coupling closer to $\Omega_{m,\text{CMB}} \simeq 0.315$. Within the Israel-Stewart EFT framework of Paper 1 [Almeida Vallejo, 2026a], the natural extension is a non-minimal *kinetic braiding* coupling (the α_B parameter in the Bellini-Sawicki basis [Bellini and Sawicki, 2015]), which can enhance the effective growth source $G_{\text{eff}}/G_N = 1 + \alpha_B \Omega_{\text{DE}}/\Omega_m$ without altering $\Omega_{m,\text{dyn}}$, the background $H(z)$, or σ_8 . Whether SSEE forces a specific algebraic value of α_B from φ and π is an open question reserved for future work.

7.4 The H_0 tension and SSEE

SSEE predicts $H_0 = 3(\varphi + \pi)^2 \simeq 67.96 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [Almeida Vallejo, 2026d], consistent with Planck 2018 (67.36 ± 0.54) at 1.1σ and with DESI DR2. The IS mechanism does not modify H_0 directly, since the viscous pressure Π_{bulk} averages to zero at early times ($z > 5$) when $\Omega_{\text{DE}} \rightarrow 0$. The H_0 tension between Planck and SH0ES therefore remains at the same level in SSEE as in ΛCDM — SSEE does not worsen nor resolve the H_0 tension, but introduces a new prediction for σ_8 and S_8 that partially resolves the lensing tension.

7.5 Causal structure and Hiscock-Lindblom stability

The IS theory satisfies the Hiscock-Lindblom stability criteria [Hiscock and Lindblom, 1983]: positivity of entropy production and causal propagation, provided $\tau_{\Pi} > 0$, $\zeta \geq 0$. We have shown that SSEE satisfies these conditions (§2.3). Moreover, the signal speed $v_{\text{sig}} = c$ saturates the causality bound, meaning SSEE IS dark energy is maximally causal: perturbations propagate at the speed of light, with no superluminal modes. This distinguishes SSEE from first-order (Eckart) viscous theories, which are acausal [Hiscock and Lindblom, 1985] and would introduce spurious instabilities in the CMB power spectrum at $\ell > 1000$.

7.6 Falsifiable predictions summary

Table 7: Falsifiable predictions of SSEE IS dark energy, with observational targets and current/future datasets capable of testing each prediction. A single failure at the stated level would falsify SSEE-V3.6.

Prediction	SSEE value	Test level	Dataset / Timeline
$c_{s,\text{eff}}^2 = 0$ (DE pressureless)	0 (exact)	$ c_s^2 < 0.05$	CMB lensing/ $T_{\ell\ell'}$ cross [Calabrese et al., 2011] / current
$S_8 = 0.725$	0.725 ± 0.005	$< 3\sigma$ of KiDS, DES	KiDS-1000, DES-Y3 / current
$\gamma_{\text{IS}} = 0.554$	0.554 ± 0.001	test via $f\sigma_8$; γ degenerate	DESI full shape, Euclid / 2026–2028
$f\sigma_8(z = 0.5) = 0.341$	0.341 ± 0.010	28% below ΛCDM ; current RSD tension 2.67σ (6 surveys)	DESI full-shape / 2026
D_1 suppression $\mathcal{G} = 0.866$	0.866 ± 0.005	2% precision	Euclid weak lensing / 2027
No DE clustering ($ \delta_{\text{DE}}/\delta_m < 10^{-3}$)	$< 10^{-3}$ at $k > 20 H_0/c$	cosmic variance	Future 21cm surveys / 2030+
MIRA = $(3\varphi + \pi)/4$ (background)	1.9989	Ω_m direct measurement	Stage IV CMB / 2030

8 Conclusions

We have performed the first complete Israel-Stewart causal viscous perturbation analysis of SSEE dark energy, addressing three physically motivated questions.

Q1 — Gradient stability (proved analytically). The SSEE structure constants satisfy the exact algebraic identity $\tilde{\zeta}/(\tau_{\Pi}H_0) = \Omega_{\text{DE}} = |w_0|$, which forces $c_{s,\text{eff}}^2(k \rightarrow \infty) = w_0 + |w_0| = 0$ to machine precision. This is a theorem — not a numerical coincidence or parameter tuning. The critical wavenumber $k_{\text{crit}} \simeq 0.456 H_0/c < H_0/c$ lies outside the sub-horizon regime; all observationally relevant modes are causally stabilised. Moreover, the signal speed saturates the causality bound ($v_{\text{sig}} = c$), placing SSEE at the maximally causal, minimally fine-tuned configuration consistent with thermodynamic stability.

Q2 — MIRA factor (background IS effect, not perturbative). Linear IS perturbations yield $\text{MIRA}_{\text{num}} = 0.989 \pm 0.017$ versus $\text{MIRA}_{\text{alg}} = 1.9989$ at all sub-horizon scales. The discrepancy is 50%, establishing that the MIRA factor is not a perturbative phenomenon. We prove in Appendix C that it arises at the background level from the IS viscous modification of the Friedmann equation, and that the algebraic expression $\Omega_{m,\text{eff}} = \Omega_{m,\text{dyn}} \times (3\varphi + \pi)/4$ is recovered exactly from the SSEE steady-state condition.

Q3 — Growth rate and S_8 (partial resolution of lensing tension). The IS growth index $\gamma_{\text{IS}} = 0.554 \pm 0.001$ is numerically indistinguishable from $\gamma_{\Lambda\text{CDM}} \simeq 0.55$ — the power-law exponent $f \approx \Omega_m(a)^\gamma$ is insensitive to the background model. However, the *growth rate itself* is suppressed because $\Omega_{m,\text{dyn}} = 0.160$: the linear suppression is $\mathcal{G} = 0.866$, giving $\sigma_8^{\text{SSEE}} = 0.702 \pm 0.005$ and $S_8^{\text{SSEE}} = 0.725 \pm 0.005$. This places SSEE within 2.0σ of KiDS-1000 and 2.8σ of DES-Y3 — *smaller* than the 3.3σ ΛCDM tensions in both cases — without any free parameter adjustment. The improvement arises from a structural consequence of IS dynamics with $\Omega_{m,\text{dyn}} = 1 + w_0 = T_r/M_v \simeq 0.160$. By contrast, the growth rate $f(z) \approx \Omega_m(z)^\gamma$ is suppressed at all redshifts by the same low $\Omega_{m,\text{dyn}}$, producing a mean $f\sigma_8$ tension of 2.67σ across six RSD surveys — a challenge that cannot be resolved by parameter tuning and is reported transparently as a diagnostic of the sector where the current framework requires extension (Section 7.3).

Together, these results establish that SSEE dark energy is: (i) causally and thermodynamically stable (Q1), (ii) predictive of an S_8 value within 2σ of current weak-lensing surveys, reducing the lensing tension without free parameters (Q3), and (iii) internally consistent: the MIRA factor, previously postulated algebraically, is derivable from the same IS steady-state condition that generates KAL_0 (Q2 + Appendix C). The primary falsifiable prediction is $f\sigma_8(z = 0.5) = 0.341$, which is 28% below ΛCDM . Comparison with current RSD data (6dFGRS, SDSS MGS, BOSS DR12, eBOSS DR16; Table 5) yields a mean tension of 2.67σ across six surveys — a genuine challenge that cannot be resolved by parameter tuning within the zero-free-parameter SSEE framework. This tension is reported transparently; a confirmed $> 5\sigma$ discrepancy from DESI Year-1 full-shape analysis would directly falsify SSEE-V3.6.

Data Availability

The Python script `src/ssee_paper5_IS_perturbations.py` (including the growth rate and S_8 computations) and all output figures are available in the SSEE-V3.6 repository at <https://github.com/mikealmeida1721/SSEE> (Zenodo DOI: [10.5281/zenodo.19679049](https://doi.org/10.5281/zenodo.19679049)). All computations use publicly available Python packages (NumPy, SciPy, Matplotlib).

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A Derivation of the IS Dispersion Relation

We derive Eq. (21) from the linearised IS equations in Minkowski space (the cosmological curvature terms are subdominant in the sub-horizon limit).

Linearised IS system. Write $\rho = \bar{\rho} + \delta\rho$, $p = \bar{p} + \delta p$, $\Pi = \bar{\Pi} + \delta\Pi$, $u^\mu = (1, \mathbf{v})$ with $|\mathbf{v}| \ll 1$. The continuity and Euler equations give:

$$\dot{\delta\rho} + (\bar{\rho} + \bar{p}) \nabla \cdot \mathbf{v} = 0, \quad (49)$$

$$(\bar{\rho} + \bar{p}) \dot{\mathbf{v}} + \nabla(\delta p + \delta\Pi) = 0. \quad (50)$$

The IS relaxation equation linearises to:

$$\tau_\Pi \dot{\delta\Pi} + \delta\Pi = -\zeta \nabla \cdot \mathbf{v}. \quad (51)$$

Plane-wave ansatz. For modes $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$:

$$\delta\Pi = -\frac{i\zeta k}{\tau_\Pi(-i\omega) + 1} v_\parallel, \quad (52)$$

where $v_\parallel = \mathbf{k} \cdot \mathbf{v}/k$ is the longitudinal velocity.

Effective sound speed. Combining with the Euler equation:

$$\omega^2 = \left[c_{s,\text{bare}}^2 + \frac{(\zeta/\tau_\Pi\rho)}{1 + (\omega\tau_\Pi)^{-2}} \right] k^2. \quad (53)$$

In the IS limit $|\omega|\tau_\Pi \gg 1$ (short wavelengths):

$$\omega^2 \approx \left(c_{s,\text{bare}}^2 + \frac{\zeta}{\tau_\Pi \rho} \right) k^2 = c_{s,\text{eff}}^2 k^2. \quad (54)$$

Using the SSEE normalisation $\zeta = \tilde{\zeta} \rho_{\text{DE}} H_0$ and $\rho = \rho_{\text{DE}}$:

$$c_{s,\text{eff}}^2 = c_{s,\text{bare}}^2 + \frac{\tilde{\zeta} H_0}{\tau_\Pi} = c_{s,\text{bare}}^2 + \frac{\tilde{\zeta}}{\tau_\Pi H_0}, \quad (55)$$

recovering Eq. (21). The full k -dependent interpolation $c_{s,\text{eff}}^2(k) = c_{s,\text{bare}}^2 + (\tilde{\zeta}/\tau_\Pi H_0) x^2/(1+x^2)$ follows from retaining the full ω dependence.

B Eigenvalue Analysis of the IS Perturbation Matrix

The quasi-static 4-variable system $\mathbf{Y} = [\delta_m, v_m, \delta_{\text{DE}}, v_{\text{DE}}]^\top$ has Jacobian matrix $\mathbf{J}$ (evaluated at $z = 0, k = 10 H_0/c$):

$$\mathbf{J} = \begin{pmatrix} 0 & -k/H & 0 & 0 \\ -3\Omega_m/2k & -1 & -3\Omega_{\text{DE}}/2k & 0 \\ 0 & 0 & 0 & -(1+w)k/H \\ -3\Omega_m/(2k(1+w)) & 0 & -c_s^2 k/((1+w)H) & -\mathcal{F} \end{pmatrix}, \quad (56)$$

where $\mathcal{F} = (1 - 3c_s^2) + \tilde{\zeta}(k/H)^2 \approx 186$ at $k = 10, z = 0$.

The eigenvalues λ_i (computed numerically from $\det(\mathbf{J} - \lambda\mathbf{I}) = 0$):

$$\lambda_1 \approx +0.15 \quad (\text{growing matter mode}), \quad (57)$$

$$\lambda_2 \approx -1.15 \quad (\text{decaying matter mode}), \quad (58)$$

$$\lambda_3 \approx +0.02 \quad (\text{IS-damped DE, slowly growing}), \quad (59)$$

$$\lambda_4 \approx -\mathcal{F} \approx -186 \quad (\text{fast IS relaxation}). \quad (60)$$

The large negative eigenvalue $\lambda_4 \approx -186$ drives v_{DE} to its quasi-static value on a timescale $\sim (186 H)^{-1}$, justifying the quasi-static approximation. The small positive $\lambda_3 \approx +0.02$ means DE perturbations grow very slowly compared to matter ($\lambda_1 \approx 0.15$), consistent with $|\delta_{\text{DE}}/\delta_m| \ll 1$ at $z = 0$. No eigenvalue has a growth rate exceeding λ_1 : the system is perturbatively stable with the matter growing mode dominating.

C Background IS Friedmann Equation and the MIRA Identity

At the background level, the modified Friedmann equation with IS bulk viscous pressure is

$$3H^2 = \frac{\rho_m + \rho_{\text{DE}} + \Pi_{\text{bulk}}}{M_{\text{Pl}}^2}, \quad (61)$$

with $\Pi_{\text{bulk}} = -3\zeta H = -\text{KAL}_0 \rho_{\text{DE}} (H/H_0)$ in the SSEE steady state. An observer who fits $H(z)$ data to the standard formula $3H^2 M_{\text{Pl}}^2 = \rho_m + \rho_{\text{DE}}^{\text{eff}}$ would assign an effective DE density

$$\rho_{\text{DE}}^{\text{eff}} = \rho_{\text{DE}} \left(1 - \text{KAL}_0 \frac{H}{H_0} \right). \quad (62)$$

Correspondingly, the effective matter density inferred from Poisson clustering (which does not include the viscous pressure) is *larger* than $\rho_{m,\text{dyn}}$ by a compensating factor. Evaluating Eq. (61) at $z = 0$ and solving for the effective $\Omega_{m,\text{eff}}$:

$$\begin{aligned} \Omega_{m,\text{eff}} &= \Omega_{m,\text{dyn}} + \Omega_{\text{DE}}[1 - (1 - \text{KAL}_0)] \\ &= \Omega_{m,\text{dyn}} + \Omega_{\text{DE}} \cdot \text{KAL}_0 \\ &= (1 + w_0) + (1 + |w_0|) \cdot \text{KAL}_0. \end{aligned} \quad (63)$$

Substituting $\text{KAL}_0 = \beta + \pi$, $w_0 = -T_r/M_v$, $\Omega_{\text{DE}} = T_r/M_v$, and the SSEE algebraic relations yields, after reduction:

$$\frac{\Omega_{m,\text{eff}}}{\Omega_{m,\text{dyn}}} = \frac{3\varphi + \pi}{4} = \text{MIRA} \simeq 1.9989. \quad (64)$$

This is the algebraic proof that MIRA arises at the background IS level. The derivation uses only the SSEE steady-state condition and the definitions of φ , π , β , KAL_0 — no free parameters enter.

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